J-C-30A
October 7, 1970
SUPERSEDING
Fed. Spec. J-C-30
June 6, 1965

FEDERAL SPECIFICATION

CABLE AND WIRE, ELECTRICAL (POWER, FIXED INSTALLATION)

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single and multiple conductor cables and wires employing soft-annealed copper or aluminum conductors insulated with rubber, thermoplastic, cross-linked-polyethylene, chlorosulphonated polyethylene, varnished cloth, asbestos, or asbestos-varnished-cloth. They are intended to be used for transmission of power in fixed type installations; also for special purpose applications (e.g. control) when appropriate.

1.2 Classification.

1.2.1 Types. Cable and wire furnished under this specification shall be of the types specified in table I.

1.3 Type designation. Cables and wires covered by this specification shall be identified by a type designation, formed as indicated below. This type designation is intended for cataloging and ordering purposes, and not for surface printing on the wire or cable.

RHW-	10-	A-	E-	4/18-	SR-	SJ-	G-	9
UL type designation (1.3.1)	Maximum working voltage (1.3.2)	Conductor material (1.3.3)	Type of conductor(s) (1.3.4)	Number and size of conductor(s) (1.3.5)	Type of insulation (1.3.6)	Outer covering (1.3.7)	Ground wire (1.3.8)	Color code of single conductor (1.3.9)

FSC 6145

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TABLE I. Cable and wire types

UL type	UL standard No.	Description	Maximum working voltage	Temp. °C.
RHW	44	Heat and moisture resistant	5000*	75
RHH	44	Heat resistant	600	90
SA	44	Silicone, heat resistant	600	90
TW	83	Moisture resistant	600	60
THWN	83	Heat and moisture resistant	600	75
THW	83	Heat and moisture resistant	600	75
THHN	83	Heat resistant, for use in dry locations	600	90
MM	719	For use in dry locations	600	60
MMC	719	For use in dry or moist and mildly corrosive locations	600	60
UF	493	For direct burial in earth	600	60
V	133	Dry locations only, general purpose	5000*	85
VD	133	Dry locations only, general purpose	5000*	85
VM	133	Dry locations only, general purpose	5000*	85
AVA	115	Dry locations only, general purpose	5000*	110
AVB	115	Dry locations only, general purpose	5000*	90
AVL	115	Dry and wet locations, general purpose	5000*	110
A	115	Dry locations only, general purpose	300	200
AA	115	Dry locations only, general purpose	300	200
AI	115	Dry locations only, general purpose	300	125
AIA	115	Dry locations only, general purpose	600	125
SE	854	Flame and moisture retardant, no mechanical protection	600	60 or 75
USE	854	Moisture resistant, no mechanical protection, for underground use	600	60 or 75
ACHH	4	Heat-resistant, armored	600	90
ACT	4	Dry locations, armored	600	60
ACL	4	Armored, lead sheath	600	75
ACHL	4	Heat resistant, armored, lead sheath	600	75
XHHW	**	Heat and moisture resistant (cross linked polyethylene, thin wall)	600	75 or 90

*Voltage rating is maximum value. Desired voltage rating should be specified (see 1.3.2).
**It is anticipated that this type will be incorporated in UL-44.

1.3.1 UL type designation. The UL (Underwriters' Laboratories) type designation shall consist of the types specified in 1.2.1.

1.3.2 Maximum working voltage. The maximum working voltage shall be designated as follows:

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03 - for use at not more than 300 volts.

06 - for use at not more than 600 volts.

10 - for use at not more than 1000 volts.

20 - for use at not more than 2000 volts.

30 - for use at not more than 3000 volts.

40 - for use at not more than 4000 volts.

50 - for use at not more than 5000 volts.

1.3.3 Conductor material. The conductor material shall be designated as follows:

A - aluminum.

C - copper.

1.3.4 Type of conductor(s). The type of conductor(s) shall be designated as follows:

E - Solid conductor (AWG 20 to AWG 8 only).

F - Concentric-lay-stranded conductor(s) of standard flexibility, conforming to type C, class B of QQ-W-343 or class B of American National Standards Institute (ANSI) Standard C7.21, as applicable.

G - Concentric-lay-stranded conductor(s) where greater flexibility is desired, conforming to type C, class C of QQ-W-343 or class C of ANSI Standard C7.21, as applicable.

H - Rope-lay-stranded conductor(s) where extreme flexibility is desired, conforming to type RC, class G of QQ-W-343.

J - Rope-lay-stranded conductor(s) where extreme flexibility is desired, conforming to type RC, class H of QQ-W-343.

K - Bunch-stranded conductor(s) where extreme flexibility in the smaller AWG sizes is desired, conforming to type B, class K of QQ-W-343.

1.3.5 Number and size of conductor(s). The number of individual conductors of the same wire size shall be designated by that number followed by a slant line and followed by a number indicating the conductor AWG size. When cables are made up of different wire sizes, each different wire size with the number of conductors shall be individually represented, with a dash separating each different wire size. A conductor larger than No. 0000 AWG shall be designated by its cross-sectional area in circular mils.

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1.3.6 Type of insulation. The type of insulation shall be designated as follows:

R - rubber

SR - synthetic rubber

NE - neoprene

T - thermoplastic

XP - cross linked polyethylene

CP - chlorosulphonated polyethylene

A - asbestos

AV - asbestos-varnished-cloth insulated

V - varnished cloth

1.3.7 Outer covering. The outer covering shall be designated as follows:

RJ - rubber jacket

SJ - synthetic rubber jacket

NJ - neoprene jacket

TJ - thermoplastic jacket

PJ - polyamide jacket

XP - cross linked polyethylene jacket

CB - cotton braid

GB - glass braid

AB - asbestos covering

FC - fibrous covering

LS - lead sheath

AL - aluminum sheath

AR - armored

UJ - unjacketed

1.3.8 Ground wire. When a ground wire is part of a cable configuration, it shall be designated by the letter "G".

1.3.9 Color code. The color code for single conductor cables and wires shall be designated by the appropriate identifying number as shown in table II. (See 3.3.1.)

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TABLE II. Color-code identification for single conductor cables and wires

Number designator	Color	Number designator	Color
0	Black	5	Green
1	Brown	6	Blue
2	Red	7	Violet (purple)
3	Orange	8	Gray (slate)
4	Yellow	9	White

NOTE: This table is not intended to signify a color sequence but only a number-color identification reference.

1.3.10 Color code for multi-conductor cables. For 2-, 3-, and 4-conductor cables, designation of color-coding will not be necessary in the type designation (see 1.3 and 1.3.9). These multi-conductors have a definite color-coding arrangement (see 3.3.2); consequently, the number of multi-conductors in the type designation (see 1.3.5) automatically signifies the color-code.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-W-343	- Wire, Electrical (Uninsulated)
PPP-B-576	- Boxes, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-591	- Boxes, Fiberboard, Wood-Cleated
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Box, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-1055	- Barrier Material, Waterproofed, Flexible

Federal Standards:

Fed. Std. No. 123	- Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 242	- Cable and Wire, Electrical (Fixed Installation).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks, as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547	- Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-C-12000	- Cable, Cord and Wire, Electric, Packaging Of.

Military Standards:

MIL-STD-129	- Marking for shipment and storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriter's Laboratories, Inc. (UL) Standards:

UL-4	- Standard for Armored Cable.
UL-44	- Standard for Rubber-Insulated Wires and Cable.
UL-83	- Standard for Thermoplastic-Insulated Wires.
UL-115	- Standard for Asbestos- and Asbestos-Varnished-Cloth-Insulated Wires.
UL-133	- Standard for Varnished-Cloth Wires and Cable.
UL-493	- Standard for Underground Feeder and Branch Circuit Cables.
UL-719	- Standard for Nonmetallic-Sheathed Cables.
UL-854	- Standard for Service Cables.

(Applications for copies should be addressed to Underwriter's Laboratories, Inc., 207 East Ohio Street, Chicago, Ill., 60611; 1285 Walt Whitman Road, Melville, L.I., New York 11746; or 1655 Scott Blvd., Santa Clara, Calif., 95050.)

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American National Standards Institute (ANSI) Standard:

C7.21 - Aluminum Conductors, Concentric-Lay-Stranded, Specification for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Cable and wire covered by this specification shall conform to the requirements of the applicable UL standards for the types specified herein (see 1.2, 1.3, and 2.2).

3.2 Fire and casualty hazards.

3.2.1 Each contractor shall submit to the contracting agency proof that the cable or wire he proposes to supply under this specification conforms to the requirements of the applicable UL Standard. The label, or listing with re-examination, of the UL may be accepted as evidence that the cable or wire conforms to the requirements.

3.2.2 In lieu of the UL label, or listing with re-examination, the contractor shall submit independent proof, satisfactory to the contracting agency, that the cable or wire conforms to the applicable requirements of the published standards including methods of tests of the applicable UL Standard.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 Color code.

3.3.1 Single conductor. The color of a single conductor cable or wire is usually arbitrarily selected for purposes of differentiating between circuits when a number of single conductor cables or wires are to be used. Consequently, the number designated in the type designation (see 1.3.9) signifies the color of a single conductor cable or wire.

3.3.2 Multi-conductor. Multi-conductor cables shall be color-coded as follows:

2-conductor - black, white.*

3-conductor - black, white, red.*

4-conductor - black, white, red, blue.

*For type SE cable, the color white is omitted for the uninsulated neutral conductor.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Unless otherwise specified, inspections by the supplier may be performed on a statistical or production lot basis to assure that the quality of manufacture is equal to the standards set by UL. (See 6.2).

4.2 Sampling, inspection, and tests. Unless otherwise specified in the invitation for bids or request for proposal, sampling, inspection, and tests shall be in accordance with the applicable UL standard (see 1.2.1).

4.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE III.

Classification of Preparation For Delivery Defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents (exterior container)	Number per container is more or less than required. Net weight exceeds requirements.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Coils (AWG sizes 20 to 8). Cable and wire in the lengths specified (see 6.1), shall be uniformly coiled and packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant and may be provided with the special knockout feature normally used in commercial practice for dispensing purposes. Close, seal and strap the container in accordance with the appendix to PPP-B-636.

5.1.1.2 Coils (AWG sizes 6 to 2). Cable and wire in the lengths specified (see 6.1), shall be uniformaly coiled and the coil shall be spirally wrapped with water-proofed flexible barrier material conforming to PPP-B-1055, and sealed.

5.1.1.3 Coils (Armored Cable). Armored cable shall be uniformly coiled in lengths specified (see 6.1), and secured with twine.

5.1.1.4 Reels. Cable and wire shall be packaged on nonreturnable wood or metal reels designed with strong flanges of a diameter that will protect the cable or wire during shipment or storage. After the cable or wire has been wound on the reel, it shall be covered with a full wrap of water-proofed flexible barrier material conforming to PPP-B-1055. The wrap shall extend between the flanges and shall be secured in place with a minimum of two metal or non-metalic straps.

5.1.2 Level B. The coils and reels shall be packaged as specified in 5.1.1 except that the fiberboard boxes, when utilized shall conform to PPP-B-636, class domestic.

5.1.3 Level C. Cable and wire shall be packaged in accordance with the manufacturers standard practice, providing that it insures protection for the product during shipment and safe delivery to its destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Coils (AWG sizes 20 to 8). Cable and cord, packaged as specified herein shall be packed in close-fitting boxes conforming to PPP-B-576, class 2; PPP-B-585, class 2 or 3; PPP-B-591, style A or B, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant or PPP-B-640, class 2.

5.2.1.2 Coils (AWG sizes 6 to 2). Cable and wire, packaged as specified herein require no further packing.

5.2.1.3 Reels. Reels of cable and wire, packaged as specified herein, shall be lagged in accordance with the level A requirements of MIL-C-12000.

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5.2.2 Level B. Coils and reels packaged as specified herein, for level A or B, require no further packing.

5.2.3 Level C. Coils and reels packaged as specified in 5.1.3 shall be packed to assure carrier acceptance and safe arrival at destination in compliance with the National Motor Freight Classification or the Uniform Freight Classification.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2,as specified (see 6.1).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.

(b) Type designation of cable or wire to be furnished (1.3).

(c) Length of cable or wire to be furnished.

(d) Responsibility for inspection, if other than specified (4.1).

(e) Basis of inspection, if other than specified (4.1.1.).

(f) Level of packagin and packing required (5.1 and 5.2).

(g) Special marking, if required (5.1.3 and 5.2.3).

(h) Color or color coding, if other than specified (3.3).

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6.2 Requirements for tests. If laboratory tests by the government are required, purchasing officers should order cable or wire for test purposes, in addition to the number of feet required for installation.

6.3 Civil agencies procurements. Cable and wire which are centrally inventory managed and procured under this specification shall be limited to the varieties set forth in Fed. Std.No. 242. The specification may be used for the procurement of additional varieties which are not centrally procured or are approved in accordance with existing regulations.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Cable-Copper

Rail:

Cable, electric, copper, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Cable, electric, copper, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115,
National Motor Freight Classification.

Cable-Aluminum

Rail:

Cable, electric, aluminum, insulated.
Carload minimum weight 30,000 pounds.

Motor:

Cable, electric, aluminum.
Truckload minimum weight 30,000 pounds, subject to Rule 115,
National Motor Freight Classification.

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Wire-Copper

Rail:

Wire, insulated, copper.
Carload minimum weight 30,000 pounds.

Motor:

Wire, insulated, copper.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Wire-Aluminum

Rail:

Wire, insulated, aluminum.
Carload minimum weight 30,000 pounds.

Motor:

Wire, insulated, aluminum.
Truckload minumum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY INTEREST:

Custodians:

Army - EL
Navy - YD
Air Force -17

Review activities:

Army - ME
Navy - AS, YD
Air Force - 80
NSA
DSA - IS

User activities:

Army - MI, MU
Navy - MC
Air Force - 11

Preparing activity:

Army - EL

CIVIL AGENCY INTEREST:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mar*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(I* *Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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